QMTS V2

CELLULAR
INTELLIGENCE
SOLUTIONS

QMTS V2

OVERVIEW

The quad modem telemetry system (QMTS) is a multi-role communications platform designed to simultaneously utilize the wireless data networks of up to four mobile network operators in a single geographic location. The central secure QRMS server aggregates the data sent over the carrier networks before sending to any number of configured client machines. Sharing the data payload across multiple networks leads to higher throughput and greater resilience. The device has an integrated router that manages the packet queue internally, with all traffic carried over secure tunnels between the QMTS and the secure server. In addition to providing secure, reliable IP transport at up to four times a single modem solution, the QMTS has built-in GPS for accurate position reporting and tracking.

KEY APPLICATIONS

The QMTS has a wide variety of applications, including:

- Backhaul of video and other surveillance traffic.
- Use as a high bandwidth wireless router.
- Remote control of mission critical assets.
- Vehicle tracking and monitoring.
- As a secure communications tool.

KEY FEATURES

The QMTS has the following key benefits:

- Higher bandwidth than using a single operator.
- Resilience from using multiple networks.
- High security of data streams.
- The QMTS uses standard data SIMs.
- Built in router, GPIO & GPS.
- Secure server can be located anywhere including user's own secure data centre.
- Secure server serves multiple QMTS devices.

KEY SPECIFICATIONS

- Input DC voltage between 9V and 28V.
- Nominal draw current 1A@12V.
- Temperature range -20°C to +55°C
    - The unit will operate in an ambient of up to +25°C with no external heatsink.
    - The unit will operate in an ambient of up to +35°C with an external passive heatsink.
    - The unit will operate in an ambient of up to +55°C with an external actively cooled heatsink.
- Up to four phone modules (*4xSMA antennas*)

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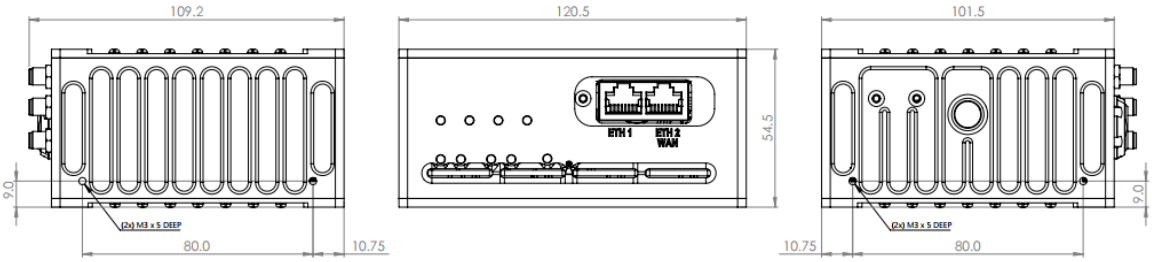
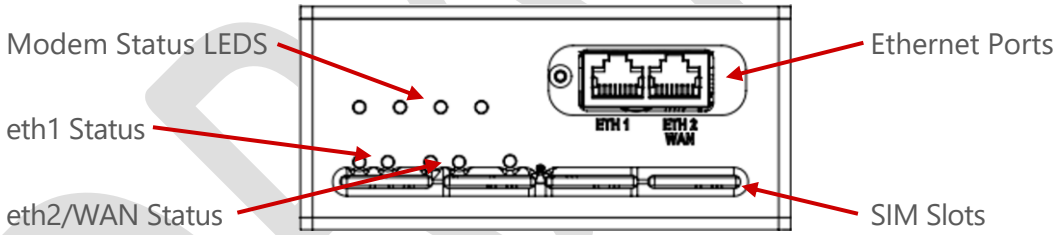
- Each phone module will support the necessary bands and RATs for the target territory.
- Four SIM slots (*micro-SIM size*).
- Two Gigabit Ethernet ports (*2 RJ45 rugged Ethernet connectors).*
- Wi-Fi transceiver for wireless network connectivity (*1xSMA antenna*).
- GPS receiver for positional and time information (*1xSMA antenna*).
- GPIO for high/low detection of anti-tamper sensor (*optional*).
- I2C bus available for more complicated sensors such as temperature or battery monitoring (*optional*).
- Dimensions (mm):

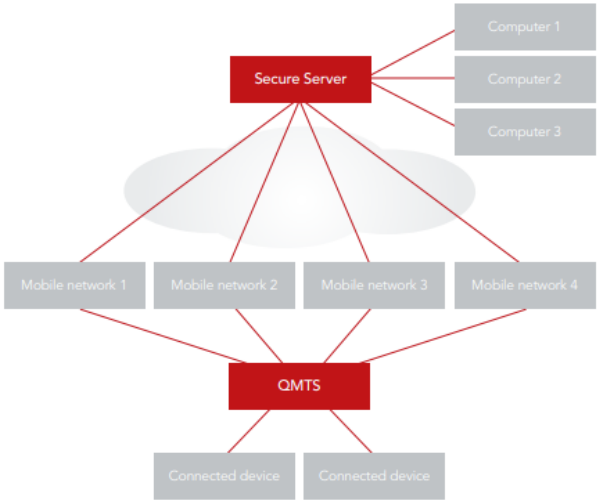


- Front Panel Details



QMTS SYSTEM ARCHITECTURE

The QMTS transmits data across separate mobile networks to the secure server. The secure server aggregates the data streams in the correct order. Traffic streams to and from the secure server over an encrypted VPN tunnel. The QMTS can connect to remote equipment, such as cameras and surveillance devices, via the two Ethernet interfaces.



USER TRAINING

- cellXion operator training and documentation provided for QMTS and QRMSSH server installation.

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